

Early Black-Hole Growth Constraints from JWST

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Abstract

Aims. Massive black holes in the early Universe must grow within limited cosmic time. We identify which JWST black-hole candidates most strongly constrain early growth models and which assumptions control these constraints. **Methods.** We compile 350 literature measurement records from 32 source families at $z \geq 4$. We analyse 220 objects with broad hydrogen-line mass estimates and include 14 additional objects in a broader comparison. We propagate reported mass uncertainties and test mass-scale shifts and alternative growth assumptions. **Results.** For $100 M_{\odot}$ seeds formed at $z = 30$, radiative efficiency $\epsilon = 0.1$, and no merger boost, 8 of the 220 objects require a lifetime-average Eddington ratio $\overline{f_{\text{Edd, req}}} > 1$. Of these, 5 exceed this threshold in at least 95% of trials sampling their quoted mass errors under fixed growth assumptions. For UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604, at least 95% of trials still require an average Eddington ratio above 1 after reducing each mass by 0.5 dex (a factor of approximately 3.2). Uncertainties in the mass estimators also affect these constraints. Delaying seeding to $z = 20$ makes GN-z11 the most demanding object. For A2744-QSO1, an independent dynamical mass estimate raises the required average Eddington ratio to approximately 1. At $\epsilon = 0.05719$, the ideal thin-disc efficiency for a non-spinning black hole, all 234 central mass estimates are reachable from $100 M_{\odot}$ seeds formed at $z = 30$ without lifetime-average accretion above the Eddington limit.

Key words. black hole physics – accretion, accretion disks – galaxies: active – galaxies: high-redshift – quasars: supermassive black holes

1 Introduction

Massive black holes observed in the early Universe must have assembled within the available cosmic time. The strength of this constraint depends on the observed black-hole mass and redshift and on assumptions about seed formation, radiative efficiency, and the duration of active accretion.

We compile a literature-based catalogue of JWST-identified high-redshift accreting black holes and candidates at $z \geq 4$, extending from the first billion years to a cosmic age of approximately 1.54 Gyr at its lower-redshift boundary. We retain published measurements, object identity links and mass-estimation sources so that measurements from different studies can be compared.

Analytic seed-growth comparisons already establish the trade-off between seed mass and sustained accretion. For example, [P. Dayal \(2024\)](#) considered light and heavy seeds at $z_{\text{seed}} = 25$ and explored a primordial-seeding interpretation. We apply the same growth equations to a larger catalogue to calculate whether each black hole could reach its observed mass while accreting, on average, at or below the Eddington limit. We test how the required rates change with seed and growth assumptions, masses drawn from published uncertainty intervals, uniform mass shifts, and alternative published measurements. For $100 M_{\odot}$ seeds formed at $z = 30$, radiative efficiency $\epsilon = 0.1$ and no mass added by mergers, 8 objects in our Balmer-line sample require an average Eddington ratio above 1. We list these objects and assess how the limitations reported for their mass estimates affect the growth constraints.

For 4 objects also studied by Dayal, we calculate separately how the required accretion rates change when we update the measured masses and redshifts and when we start growth earlier. For A2744-QSO1, we compare the catalogue’s virial mass with an independent dynamical estimate.

The contributing studies include the JADES broad-line census (I. Juodžbalis et al. 2026a), which measures current accretion and host scaling relations, and the narrow-line census (J. Scholtz et al. 2025), which identifies candidates for obscured activity. The studies targeted different kinds of objects and reached different detection limits. Their combined probability of selecting and detecting a black hole is unknown. The source-family inventory is given in Appendix A.

1.1 Theoretical framework

We use the luminosity-based Eddington ratio $f_{\text{Edd}} = L_{\text{bol}}/L_{\text{Edd}}$, where L_{bol} is the total luminosity of the accretion flow integrated over all wavelengths and L_{Edd} is the luminosity at which outward radiation force balances gravity for electron scattering in ionized hydrogen. The fixed radiative efficiency ϵ is the fraction of inflowing rest-mass energy emitted as radiation; the fraction $1 - \epsilon$ is retained by the black hole in this model. We define the available growth interval Δt and the lifetime-average Eddington ratio $\overline{f_{\text{Edd}}}$ as

$$\Delta t = t(z_{\text{obs}}) - t(z_{\text{seed}}) > 0, \quad \overline{f_{\text{Edd}}} = \frac{1}{\Delta t} \int_{t(z_{\text{seed}})}^{t(z_{\text{obs}})} f_{\text{Edd}}(t) dt. \quad (1)$$

Here $t(z)$ is the age of the Universe at redshift z , z_{seed} is the redshift at which the seed forms and growth begins, and z_{obs} is the observed redshift. The integration variable t is cosmic time, and $f_{\text{Edd}}(t)$ is the instantaneous Eddington ratio; its average includes both active and quiet phases throughout Δt . We use the exponential growth relation adopted by P. Dayal (2024), expressed in terms of the time-averaged Eddington ratio, and introduce a multiplier B_{merge} to approximate additional mass from mergers:

$$M_{\text{BH}}(t_{\text{obs}}) = B_{\text{merge}} M_{\text{seed}} \exp \left[\frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \Delta t}{t_{\text{Edd}}} \right], \quad (2)$$

Here $M_{\text{BH}}(t_{\text{obs}})$ is the black-hole mass at the observation time $t_{\text{obs}} = t(z_{\text{obs}})$, and M_{seed} is the initial seed mass. The Eddington timescale, $t_{\text{Edd}} = M_{\text{BH}} c^2 / L_{\text{Edd}} \simeq 0.45$ Gyr, is the time needed to radiate an energy equal to the black hole’s rest-mass energy at its Eddington luminosity, with c denoting the speed of light. Because the Eddington luminosity is proportional to black-hole mass, $t_{\text{Edd}} \simeq 0.45$ Gyr is the same for all black-hole masses under this assumption. The mass-growth e-folding time also depends on efficiency and accretion rate: for constant $f_{\text{Edd}} > 0$, it is $[\epsilon/(1 - \epsilon)] t_{\text{Edd}} / f_{\text{Edd}}$, or 0.05 Gyr for $\epsilon = 0.1$ and $f_{\text{Edd}} = 1$. The dimensionless merger multiplier is $B_{\text{merge}} = 1$ for the reference calculation and 2 for a factor-of-two mass boost. We represent mergers by multiplying the seed mass by a constant factor before calculating accretion. This approximation omits the timing and sequence of mergers, their mass ratios, and mass lost through gravitational radiation. The required lifetime-average ratio $\overline{f_{\text{Edd}}}_{\text{req}}$ is the average needed to reach the observed mass M_{BH} from the assumed seed within Δt :

$$\overline{f_{\text{Edd}}}_{\text{req}} = \max \left[0, \frac{\epsilon}{1 - \epsilon} \frac{t_{\text{Edd}}}{\Delta t} \ln \left(\frac{M_{\text{BH}}}{B_{\text{merge}} M_{\text{seed}}} \right) \right]. \quad (3)$$

If $B_{\text{merge}} M_{\text{seed}}$ exceeds the observed black-hole mass, the logarithm is negative and we set the required average Eddington ratio to zero. Even with zero accretion, this seed and merger contribution predict a mass larger than observed. Reproducing the observed mass would require a smaller seed, a smaller merger contribution, or a model that includes mass loss. This requirement averages accretion over the full growth interval. Inferring a duty cycle also requires assumptions about the rates during active and quiet phases.

We calculate cosmic ages using the present-day Hubble expansion rate $H_0 = 67.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and matter density parameter $\Omega_m = 0.315$, matching these 2 values in P. Dayal (2024), who cite Planck.

We impose spatial flatness and set $\Omega_\Lambda = 1 - \Omega_m = 0.685$, where Ω_Λ is the density parameter of the cosmological constant; both density parameters are relative to the present-day critical density. The age calculation includes matter and a cosmological constant and neglects radiation. Appendix B gives the supporting relations, implementation, and inverse seed-mass diagnostic. Equations 2–3 assume constant radiative efficiency. The ideal thin-disc efficiencies used in Figure 1 are 0.03775, 0.05719, and 0.42265 for dimensionless spins $a = -1, 0, +1$, respectively (J. M. Bardeen et al. 1972). Here a is the black-hole spin parameter: $a = -1$ and $+1$ denote maximal spins counter-aligned and aligned with the disc’s angular momentum, respectively, and $a = 0$ denotes a non-spinning black hole. Each track holds efficiency constant; spin evolution is unspecified.

2 Catalogue construction

We compile 350 literature measurements from 32 source families at $z \geq 4$. Each adopted mass is linked to its source, method, and reported uncertainty. We select 1 published measurement per object using the source assessments and retain the alternatives for sensitivity tests.

Growth analysis requires a numerical mass and redshift, an identified mass method, and a resolved lensing treatment. Upper limits, indirect model ranges, and masses inferred by assuming an Eddington ratio are excluded from numerical inference. We also exclude 6 catalogue entries in 3 groups where it remains unclear whether entries from different studies refer to the same physical object. Three of these entries have eligible masses. This leaves 220 primary objects and an inclusive exploratory sample of 234 objects.

For the main comparison, we use masses estimated from broad H α or H β emission lines in objects with secure or probable evidence for an accreting black hole, using the evidence classifications recorded from the source papers. Seven masses are explicitly flagged as conditional on a broad-line-region interpretation: 6 candidates with intermediate-width H α emission and possible outflow contamination (W. Ren et al. 2025), and RUBIES-EGS-49140 (A. J. Taylor et al. 2025), for which an alternative non-AGN interpretation is recorded. These 7 objects enter the expanded sample only. After the identity exclusions, the main comparison contains 220 objects, labelled primary in the tables and figures. Other limitations of a virial mass, including possible scattering, are recorded separately. COSMOS3D-13852 remains in the main sample on the basis of its published broad-line AGN identification; we test the reported scattering bias in Section 5.2. These single-epoch virial estimators combine the line width, which estimates gas velocity, with a luminosity-based estimate of the broad-line region’s radius. Their calibration is based on reverberation mapping of nearby AGN (Y. Shen 2013). Using the same line family and mass-estimation framework reduces differences between methods in the comparison. Applying these calibrations at high redshift still requires assessing gas dynamics, geometry, obscuration, and lensing for each object. We also analyse an expanded sample of 234 objects, labelled exploratory in the tables and figures. It includes the 220 primary objects, 13 additional candidates, and GN-z11. GN-z11’s mass is inferred from rest-frame ultraviolet emission lines using local virial relations (R. Maiolino et al. 2024). We include it in the expanded comparison because its lines and calibration differ from those of the Balmer-line sample. The sample division reflects the evidence for accretion and the methods being compared; the reliability of each mass estimate requires an individual assessment. When a paper gives a mass without an uncertainty, we can calculate the average Eddington ratio needed to reach that mass. We cannot calculate the probability that this ratio exceeds 1 from the published information, because no mass uncertainty distribution is available. We therefore include these objects in calculations at their published masses and omit them from probability calculations. The mass estimates retained in each sample still use different calibrations, and the contributing studies have different target-selection criteria and detection limits. Appendix A gives the source inventory, identity decisions, and sample accounting. Table 1 summarizes the catalogue and analysis sample sizes.

Table 1: Catalogue size and analysis samples.

Product or subset	Rows or objects
Measurements / provisional objects / hosts	350 / 338 / 337
Growth-eligible measurements / objects	244 / 237
Method-eligible primary measurements / objects	234 / 227
Manuscript primary / exploratory objects	220 / 234
Identity-excluded object records	6
Catalogue-only object records	101

3 Growth-model comparison

3.1 Reference model

We calculate the required lifetime-average Eddington ratio using Eq. 3, with $M_{\text{seed}} = 10^2 M_{\odot}$, $z_{\text{seed}} = 30$, $\epsilon = 0.1$, and $B_{\text{merge}} = 1$ as the reference history. We adopt $z_{\text{seed}} = 30$ as an early starting point that allows more time for light-seed growth, and test how later onset changes the required accretion rates. We rank objects by this required average Eddington ratio. Figure 1 illustrates tracks at 4 fixed efficiencies; tracks are selected for proximity to the observed masses. Appendix B.5 specifies the display selection.

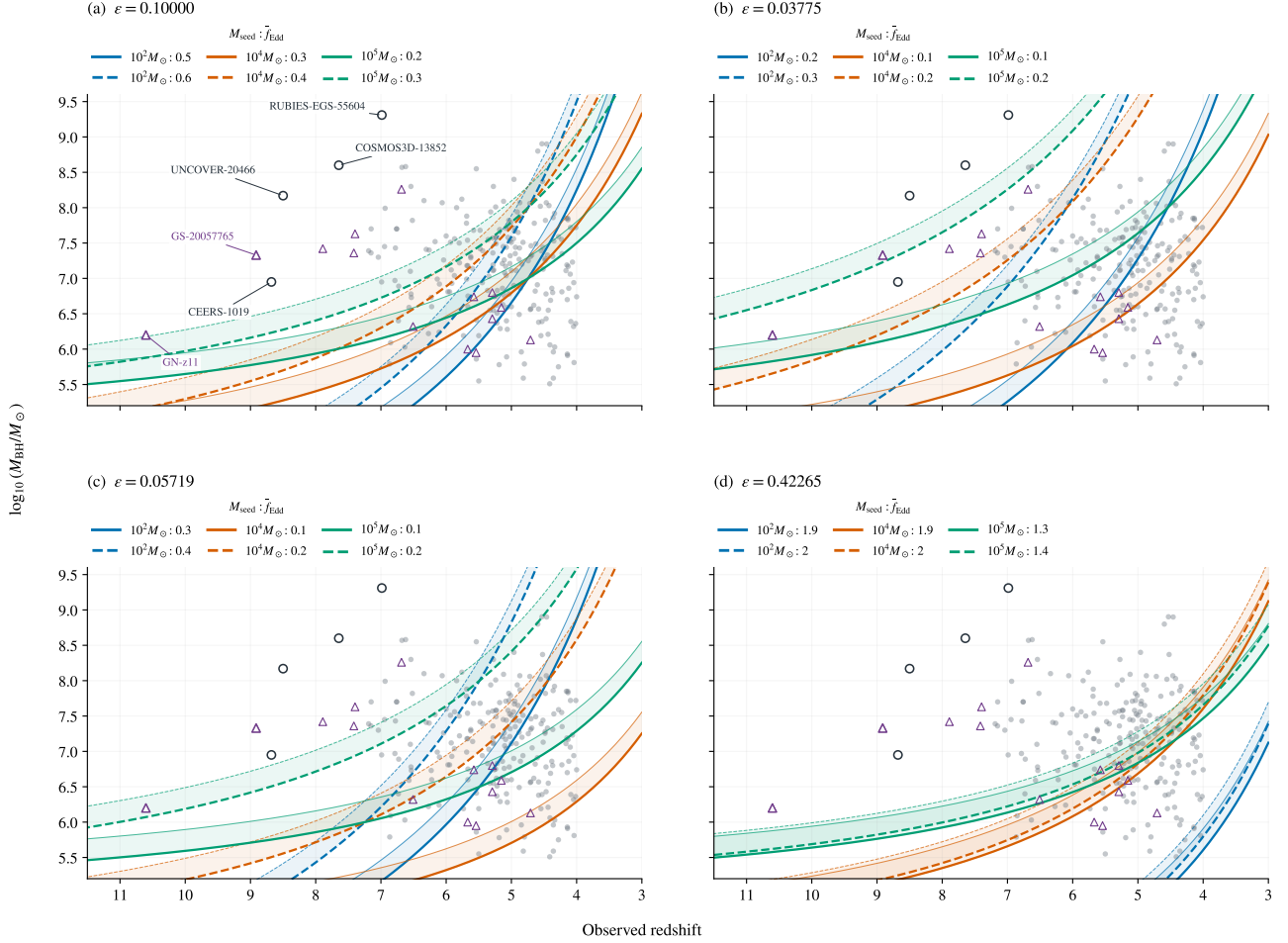
3.2 Uncertainty and sensitivity

We use Monte Carlo sampling to propagate reported asymmetric log-mass errors, with 10,000 draws per object. Each draw has equal probability of falling above or below the central mass, with its distance drawn from a half-normal distribution scaled by the corresponding upper or lower error. Redshifts are held fixed. For each sampled mass, we calculate the average Eddington ratio needed to grow from the assumed seed to that mass. The fraction of the 10,000 Monte Carlo draws with a required ratio above 1 estimates the probability that growth requires an average rate above the Eddington limit, given the adopted mass-error distribution and growth assumptions. Probability calculations include 208 primary and 222 expanded-sample objects; the 12 objects without reported errors enter only calculations at their published masses.

We test mass-scale sensitivity by shifting all adopted log masses and their error draws by offsets of -0.5 , -0.3 , 0 , $+0.3$, and $+0.5$ dex, keeping the sample and growth assumptions fixed. These fixed offsets are stress tests at the uncertainty scale of single-epoch mass estimates (Y. Shen 2013). For each offset, we report the fraction of shifted Monte Carlo draws that require an average Eddington ratio above 1. Each probability assumes that particular mass shift and the stated seed mass, starting redshift, radiative efficiency, and merger contribution. Sampling and conversion details are given in Appendix B.5.

Growth tracks across radiative-efficiency assumptions

$z_{\text{seed}} = 30$; bands span $B = 1$ (thick edge) to $B = 2$ (thin edge). Seed mass is encoded by colour; exact rates are listed above each panel.



Grey circles: primary (220); purple triangles: exploratory only (14). Outlined targets are named in panel (a).

Figure 1: Growth tracks with 220 primary and 14 exploratory-only objects. Panels show 4 constant radiative efficiencies at $z_{\text{seed}} = 30$; colours distinguish seeds of 10^2 , 10^4 , and $10^5 M_{\odot}$, and legends list the selected average Eddington ratios. Shaded bands span merger boosts $B = 1$ – 2 , with thick and thin edges respectively. The bands show the effect of the assumed merger boost. Six named objects are outlined in every panel. The reversed redshift axis spans $z = 11.5$ to $z = 3$. Excluded identities and objects without eligible masses are omitted.

4 Results

4.1 Required accretion under the reference assumptions

For a $10^2 M_{\odot}$ seed under the reference history, the median required lifetime-average Eddington ratio across the 220 primary objects is $\overline{f_{\text{Edd, req}}} = 0.573$ (interquartile range 0.485–0.673), with 8 point requirements above unity and none above 2. Across the inclusive 234-object sample, the corresponding median is 0.574 (interquartile range 0.488–0.679), with 14 above unity. These sample statistics use each object’s point requirement. UNCOVER-20466 is most demanding: its point requirement is 1.452, while the median required lifetime-average Eddington ratio over its mass-error draws is 1.453 (16th–84th percentiles 1.354–1.549). Table 2 lists the 8 primary objects above the threshold; their ordering by point requirement differs from ordering by quoted-error probability.

The shorter growth time at $z = 10.603$ places GN-z11 second in the required-rate ranking despite its smaller mass. Its UV virial estimator places it in the exploratory sample.

Table 2: The 8 primary objects above the reference threshold, ordered by their point requirements. Masses and errors are in $\log_{10}(M/M_{\odot})$; all estimators are single-epoch virial. f_0 is the required lifetime-average Eddington ratio evaluated at the adopted central mass under the reference history; f_{16} is the 16th percentile of this required ratio over the object’s mass-error draws, and $P_0 = P(f_{\text{Edd,req}} > 1)$. f_- and P_- apply the fixed -0.5 dex offset. Δm (dex) is the reduction in log mass that brings the central required average Eddington ratio to 1, with the growth assumptions unchanged. All probabilities use the quoted mass errors and fixed growth assumptions. Additional mass-estimator limitations are in Table 7.

Object/source	Line	Log mass	f_0	f_{16}	P_0	f_-	P_-	Δm
UNCOVER-20466 J. E. Greene et al. (2024)	H β	$8.17^{+0.42}_{-0.42}$	1.452	1.354	> 0.999	1.335	> 0.999	1.921
COSMOS3D-13852 X. Lin et al. (2026)	H β	$8.60^{+0.35}_{-0.33}$	1.314	1.247	> 0.999	1.214	> 0.999	1.576
RUBIES-EGS-55604 ^a D. D. Kocevski et al. (2025)	H β	$9.31^{+0.10}_{-0.10}$	1.267	1.250	> 0.999	1.180	> 0.999	1.541
CEERS-1019 R. L. Larson et al. (2023)	H β	$6.95^{+0.37}_{-0.37}$	1.205	1.115	0.988	1.083	0.822	0.842
GN-38509 ^b I. Juodžbalis et al. (2026a)	H α	$8.57^{+0.37}_{-0.38}$	1.065	1.005	0.860	0.984	0.392	0.398
J0910_2028_12910 ^a C. J. Baccus & X. Xu (2026)	H α	$8.58^{+0.01}_{-0.01}$	1.053	1.051	> 0.999	0.973	< 0.001	0.330
CEERS-00717 Y. Harikane et al. (2023)	H α	$7.99^{+0.16}_{-0.17}$	1.027	0.998	0.821	0.942	0.018	0.160
ZS7 ^c H. Übler et al. (2024)	H β	$7.70^{+0.40}_{-0.40}$	1.023	0.954	0.625	0.934	0.173	0.130

^a A separate 0.5-dex calibration scale is reported for RUBIES-EGS-55604 and J0910_2028_12910; these scales are omitted from the quoted errors and probabilities shown here. ^b GN-38509’s quoted mass errors already include 0.3-dex calibration scatter. ^c ZS7’s quoted 0.4-dex error already includes calibration scatter. Unmarked rows have no separate numerical calibration scale recorded; their quoted errors may omit other mass-estimator uncertainties.

4.2 Effect of reported mass uncertainties

Of the 8 primary point requirements above unity, 6 remain above it at the 16th percentile and 5 have conditional $P(f_{\text{Edd,req}} > 1) \geq 0.95$ (Fig. 2). In the exploratory sample, 14 objects have a median required lifetime-average Eddington ratio above unity, 10 exceed unity at the 16th percentile, and 8 have $P \geq 0.95$; 15 exceed unity at the 84th percentile. Here each median and percentile is taken over the individual object’s mass-error draws. Probabilities follow Section 3.2 and use 208 primary and 222 exploratory objects with reported errors. The 12 primary objects without errors enter only point counts.

Objects above the reference growth threshold

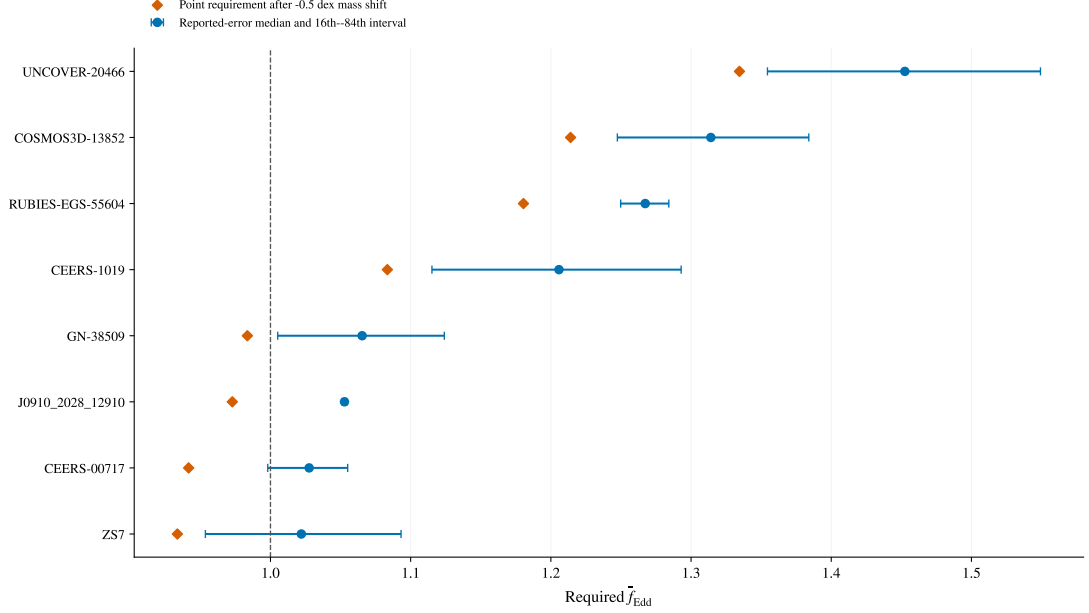


Figure 2: The 8 primary objects whose reference point requirements exceed unity. Circles and bars show medians and 16th–84th percentile intervals of the required lifetime-average Eddington ratio over each object’s mass-error draws; diamonds show point requirements after a -0.5 dex mass shift. The vertical line is the reference threshold. The selected objects all have reported errors; these intervals do not include additional estimator systematics.

4.3 Sensitivity to the virial-mass scale

Downward mass shifts of 0.3 and 0.5 dex reduce the primary point/16th-percentile/ $P \geq 0.95$ counts from 8/6/5 to 6/5/4 and 4/3/3, respectively (Table 3). UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604 retain $P \geq 0.95$ at -0.5 dex. Biases specific to each source may exceed this offset (Section 5.2), and the required rates still depend on seed mass, formation time, and radiative efficiency.

Table 3: Fixed mass-scale stress tests under the reference growth history. Triplets give counts above unity at the point estimate, at the 16th percentile, and with conditional $P(\bar{f}_{\text{Edd, req}} > 1) \geq 0.95$. The probability-based counts use 208 primary and 222 exploratory objects with reported errors; point counts use all 220 and 234 objects.

Mass offset (dex)	Primary	Exploratory
-0.5	4 / 3 / 3	7 / 5 / 4
-0.3	6 / 5 / 4	10 / 7 / 5
0	8 / 6 / 5	14 / 10 / 8
$+0.3$	11 / 11 / 10	17 / 15 / 14
$+0.5$	15 / 12 / 11	21 / 18 / 15

Replacing the redshifts, masses, and errors of 44 catalogue objects matched to the Baccus & Xu sample by identifier and confirmed by position with published Table 1 values (C. J. Baccus & X. Xu 2026) preserves the primary 8/6/5 and exploratory 14/10/8 counts and both top-five rankings (Table 4). Of 5 unmatched records, 1 is identity-excluded; retaining or omitting the other 4 leaves these results unchanged (216/230 objects when omitted). The test updates the matched measurements while keeping all 6 identity

exclusions, method classes, growth parameters, and sampling rules fixed.

Table 4: Effect of adopting the published Baccus measurements after excluding unresolved identities. Every row excludes all open identity groups. Counts denote point requirements above unity, 16th percentiles above unity, and conditional $P \geq 0.95$, respectively. The adopted catalogue values define the baseline.

Sample	Measurement treatment	Objects	Counts
Primary	Frozen values	220	8/6/5
Primary	Published; retain unmatched	220	8/6/5
Primary	Published; omit unmatched	216	8/6/5
Exploratory	Frozen values	234	14/10/8
Exploratory	Published; retain unmatched	234	14/10/8
Exploratory	Published; omit unmatched	230	14/10/8

4.4 Model compatibility and measurement choice

Increasing the seed mass to $10^3 M_\odot$ reduces the primary count above unity from 8 to 3; $10^5 M_\odot$ seeds remove all point exceedances. Delaying seeding to $z = 20$ raises the primary count to 18, while $\epsilon = 0.05719$ removes all point exceedances in both samples (Table 5). Each test changes 1 assumption: seed mass (P. Dayal 2024), available growth time, or the efficiency of an ideal thin disc around a non-spinning black hole (J. M. Bardeen et al. 1972).

Table 5: Point-estimate requirements above $\overline{f_{\text{Edd}}} = 1$ under fixed histories. Each row changes only the named parameter from the reference ($M_{\text{seed}} = 10^2 M_\odot$, $z_{\text{seed}} = 30$, $\epsilon = 0.1$, $B_{\text{merge}} = 1$). Counts use central mass estimates and provisional object identities.

Scenario	Primary (220)	Exploratory (234)
Reference	8	14
$M_{\text{seed}} = 10^3 M_\odot$	3	5
$M_{\text{seed}} = 10^5 M_\odot$	0	0
$z_{\text{seed}} = 20$	18	24
$\epsilon = 1 - \sqrt{8/9}$	0	0

Removing the 6 ambiguous records from the 223/237-object sets before identity exclusion preserves threshold counts and top-five rankings in every listed scenario, including the reference probability counts. The median required lifetime-average Eddington ratio across primary objects, evaluated using their central masses, changes from 0.571 to 0.573; the objects above the reference threshold remain the same.

Figure 3 shows the maximum fixed efficiency allowing $\overline{f_{\text{Edd}_{\text{req}}}} \leq 1$ for 5 targets as a function of seed mass. The $z_{\text{seed}} = 20$ and 30 curves show GN-z11’s greater sensitivity to the starting time. These boundaries use central mass estimates and assume no merger boost.

Appendix C gives the fractions of objects whose masses can be reproduced from each seed-mass range under specified accretion rates, spin-dependent efficiencies, and merger contributions.

Seven alternate-measurement comparisons across 6 objects (2 for CEERS-2782) span mass shifts of -0.600 to $+0.949$ dex and required- f_{Edd} shifts of -0.083 to $+0.121$ (Fig. 4). Both estimates remain below unity in every pair. This comparison covers 6 objects below the threshold. Testing the objects above it requires additional independent mass estimates.

Seed mass and radiative-efficiency constraints

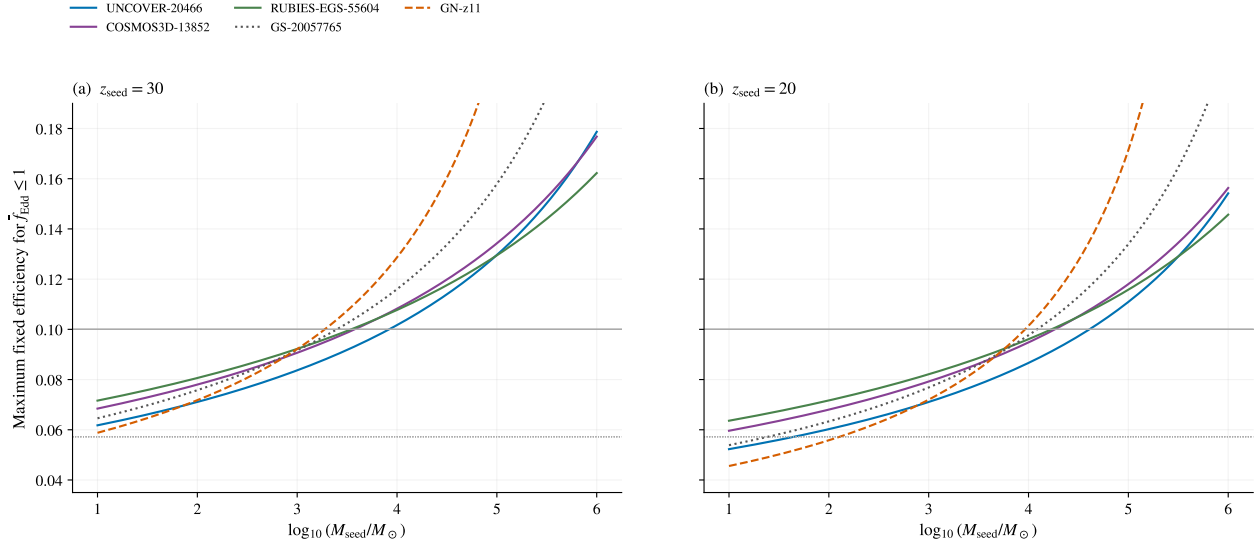


Figure 3: Seed-mass-efficiency boundaries for Eddington-limited average growth at 2 starting redshifts. Each curve uses an adopted point mass and no merger boost. Below a curve the required average is at most unity. Horizontal guides mark $\epsilon = 0.1$ (solid grey) and $1 - \sqrt{8/9}$ (dotted grey). GN-z11 (dashed) and GS-20057765 (dotted) are exploratory; the other objects are primary. Each curve holds efficiency constant.

Growth requirements from alternate mass estimates

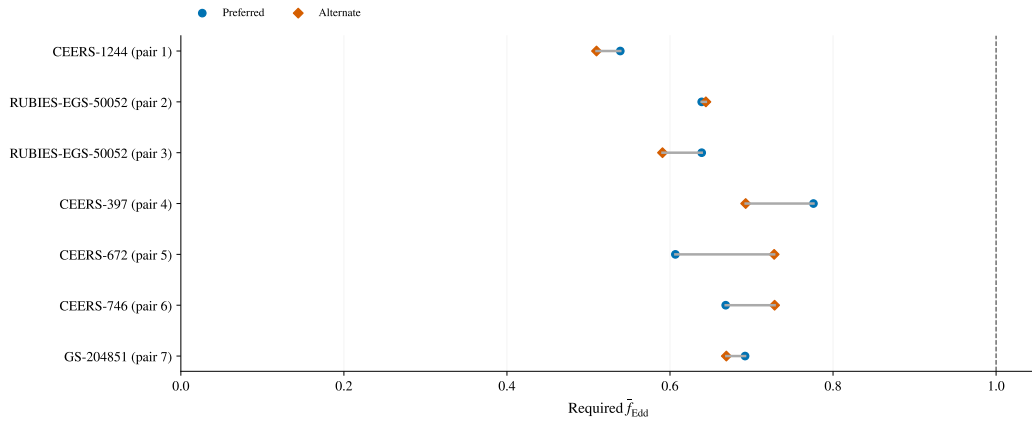


Figure 4: Preferred (circles) and alternate (diamonds) point requirements for 7 measurement pairs across 6 primary objects. Pair numbers follow the released alternate-measurement table, which supplies stable measurement IDs; RUBIES-EGS-50052 (CEERS-2782) has 2 alternatives. All pairs remain below the reference threshold.

4.5 Separating measurement and starting-time changes

For 4 unambiguous matches to Table 1 of [P. Dayal \(2024\)](#), Table 6 first replaces the earlier masses and redshifts with our adopted values at $z_{\text{seed}} = 25$, then changes seeding to $z = 30$. All columns use our cosmology, $\epsilon = 0.1$, $10^2 M_\odot$ seeds, and no merger boost. Earlier seeding lowers the requirements more than the measurement updates change them. The comparison isolates these 2 effects for the 4 matched objects using our growth calculation.

Table 6: Measurement and starting-time comparison for 4 objects shared with the Dayal study. Columns give required average f_{Edd} ; the first 2 share $z_{\text{seed}} = 25$. The analysis files provide the input measurements and their source locations.

Object	Earlier inputs, 25	Adopted inputs, 25	Adopted inputs, 30
GN-z11	1.577	1.578	1.437
A2744-QSO1	0.974	0.974	0.929
UNCOVER-20466	1.540	1.548	1.452
CEERS-00717	1.074	1.076	1.027

4.6 External dynamical-mass check: A2744-QSO1

For A2744-QSO1, we compare the virial mass adopted in the catalogue $\log_{10}(M/M_\odot) = 7.3 \pm 0.2$ ([J. E. Greene et al. 2024](#)) with the independent MOKA3D dynamical estimate 7.7 ± 0.3 ([I. Juodžbalis et al. 2026b](#)), using $z = 7.04$ and the reference history. We retain the values from [arXiv:2508.21748v2](#); the reference lists the subsequent journal publication. The dynamical estimate depends on lens reconstruction and inclination. We propagate the quoted two-sided mass error; the separate inclination-free lower limit is a one-sided constraint. Using the central mass estimates, the required average Eddington ratio rises from 0.929 to 1.000. With a symmetric normal approximation to the quoted log-mass errors, the exact 16th–84th percentile intervals are 0.895–0.964 and 0.947–1.052, respectively. For the dynamical mass, the unrounded required ratio is just below 1. Its uncertainty interval includes average accretion rates both below and above the Eddington limit. With this mass-error distribution and the growth assumptions held fixed, the probability that the required average Eddington ratio exceeds 1 is 0.497. This comparison tests the mass of 1 catalogue object. Calibrating the virial sample would require independent measurements for more objects.

5 Discussion

5.1 Dependence on seed and growth assumptions

The primary objects above the reference threshold lie at $z = 6.6214$ – 8.679 ; the corresponding range for the inclusive exploratory sample is $z = 6.6214$ – 10.603 . Most catalogue objects lie at lower redshift and serve as the comparison sample. Their growth requirements constrain combinations of seed mass, starting time, efficiency, and accretion. Several formation histories satisfy these constraints.

At fixed seed, growth interval, and merger factor, required f_{Edd} scales as $\epsilon/(1 - \epsilon)$. Lowering ϵ from 0.1 to 0.05719 multiplies all requirements by 0.54594: the largest becomes 0.793. The requirement for super-Eddington growth therefore depends on the assumed efficiency. Larger seed masses, earlier growth onset, and merger contributions also reduce the required accretion rate.

5.2 Mass-estimator limitations

Table 7 summarizes the mass-estimator limitations for the 8 primary objects above the reference threshold.

Table 7: Mass-estimator limitations for all 8 primary objects above the reference threshold. Caveats summarize the source records cited in Table 2.

Object	Mass-estimator caveat
UNCOVER-20466	Hbeta; mass depends on lens model and BLR interpretation.
COSMOS3D-13852	Hbeta; source warns of possible 1–2 dex scattering bias.
RUBIES-EGS-55604	Hbeta; dust-corrected mass; separate 0.5 dex calibration scale.
CEERS-1019	Hbeta; broad component has comparatively low significance.
GN-38509	Halp; quoted mass errors include 0.3 dex calibration scatter.
J0910_2028_12910	Halp; 0.01 dex formal error excludes 0.5 dex mass scale.
CEERS-00717	Halp; local calibration; no numeric systematic supplied.
ZS7	Hbeta; quoted 0.4 dex already includes calibration scatter.

Some mass-estimator biases may exceed the -0.5 dex shift tested above. UNCOVER-20466’s demagnified $H\beta$ mass depends on lensing and broad-line interpretation (J. E. Greene et al. 2024); RUBIES-EGS-55604’s dust-corrected $H\beta$ mass has a separately recorded 0.5 dex calibration scale (D. D. Kocevski et al. 2025). For COSMOS3D-13852, scattering could inflate the $H\beta$ mass by 1–2 dex (X. Lin et al. 2026). Applying downward offsets of 1 and 2 dex gives point requirements of 1.115 and 0.916, respectively. These stress tests use the bias range discussed by the source; the actual correction remains uncertain.

The mass reductions in Table 2 show how far each central mass lies above the value reachable at an average Eddington ratio of 1. We subtract the log mass predicted by Eq. 2 at $f_{\text{Edd}} = 1$ from the observed log mass, keeping the seed, starting redshift, efficiency, and merger contribution fixed. For J0910_2028_12910, a reduction of 0.330 dex is sufficient. This is smaller than its separately reported 0.5-dex calibration scale (C. J. Baccus & X. Xu 2026), although its 0.01-dex quoted error gives $P > 0.999$ in the reported-error calculation. COSMOS3D-13852 requires a reduction of 1.576 dex, within the 1–2 dex range of possible scattering bias. UNCOVER-20466 requires 1.921 dex; its lensing and broad-line assumptions still require independent tests. ZS7’s quoted 0.4 dex already includes calibration scatter (H. Übler et al. 2024). The probabilities therefore reflect different combinations of measurement and calibration uncertainty across the sample.

Confirming the weak broad $H\beta$ component in CEERS-1019 would strengthen the basis of its virial mass (R. L. Larson et al. 2023), although calibration uncertainty would remain. An independent mass estimate for J0910_2028_12910 could test its 0.330-dex margin above the threshold. Shared calibration errors would limit how strongly agreement between estimators supports the inferred mass.

5.3 Growth constraints for individual objects

UNCOVER-20466 and GS-20057765 have reference requirements of 1.452 and 1.355, with 16th percentiles above unity. Their point requirements remain above unity for $10^3 M_\odot$ seeds (1.217 and 1.101), but fall to 0.746 and 0.592 for $10^5 M_\odot$ seeds. At $\epsilon = 0.05719$, light seeds require 0.793 and 0.740. For Eddington-limited active phases separated by zero-accretion phases, these imply active fractions of about 79 and 74 per cent of the available time, requiring a sustained fuel supply.

GN-z11’s reference requirement is 1.437. Seeding at $z = 20$ raises it to 1.880, above UNCOVER-20466’s 1.733, reversing their order. A mass measured with a method comparable to the Balmer-line sample would test this timing constraint. Appendix D shows the joint dependence on seed mass and starting redshift

for GN-z11 and UNCOVER-20466. GN-z11 meets the primary evidence criterion and is placed in the exploratory sample because it uses a UV mass estimator.

5.4 Limitations

The contributing studies use different selection criteria and detection limits. Inferring population demographics would require a combined completeness model. Virial calibration and geometry errors are incompletely reported. The common mass offsets test a uniform shift; biases specific to individual sources, correlated errors, and selection-dependent scatter remain unmodelled.

Two duplicate pairs have been reconciled using the source papers; the identities of 3 groups remain unresolved (Appendix A.1). Excluding all 6 affected records preserves the threshold results. Reinstating these records requires checking their spectra, masses, images, or apertures against the source papers. Counts of unique objects and hosts remain provisional.

Equation 2 describes growth through an average accretion rate, a fixed merger multiplier, and constant radiative efficiency. Feedback and individual accretion histories remain unspecified. The probabilities hold these assumptions fixed. Averaging over them would require justified parameter distributions and correlations; identifying seed origins would also require formation models and population information.

Independent source checks cover all admitted mass/error fields and central redshifts. Coordinate checks and missing fields are documented with the release. Validation of redshift errors, full posteriors, and auxiliary observables is incomplete. The catalogue covers the declared sources through 3 September 2026; other JWST discoveries may fall outside this source list.

6 Conclusions

For the adopted masses and growth assumptions, we find:

- For $100 M_{\odot}$ seeds formed at $z = 30$, $\epsilon = 0.1$, and no merger contribution, 8 of the 220 objects in the main sample require an average Eddington ratio above 1. Five do so in at least 95% of Monte Carlo draws from their quoted mass-error distributions.
- For UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604, at least 95% of draws still require an average Eddington ratio above 1 after reducing the masses by 0.5 dex. Their mass estimates require tests of lensing, scattering, and calibration, respectively.
- The assumed start of growth changes which objects require average accretion above the Eddington limit and which requires the highest rate. At constant efficiency $\epsilon = 0.05719$, every central mass in the 234-object expanded sample is reachable from a $100 M_{\odot}$ seed at $z = 30$ with an average Eddington ratio below 1.
- For A2744-QSO1, replacing the adopted virial mass with the independent dynamical estimate raises the required average Eddington ratio from 0.929 to approximately 1. Its mass uncertainty allows average rates both below and above the Eddington limit.

Excluding the 6 records with unresolved identities preserves the threshold counts. Counts of unique objects in the full catalogue remain provisional.

Appendix A Source-family inventory

Manuscript sample accounting

338 catalogue object records

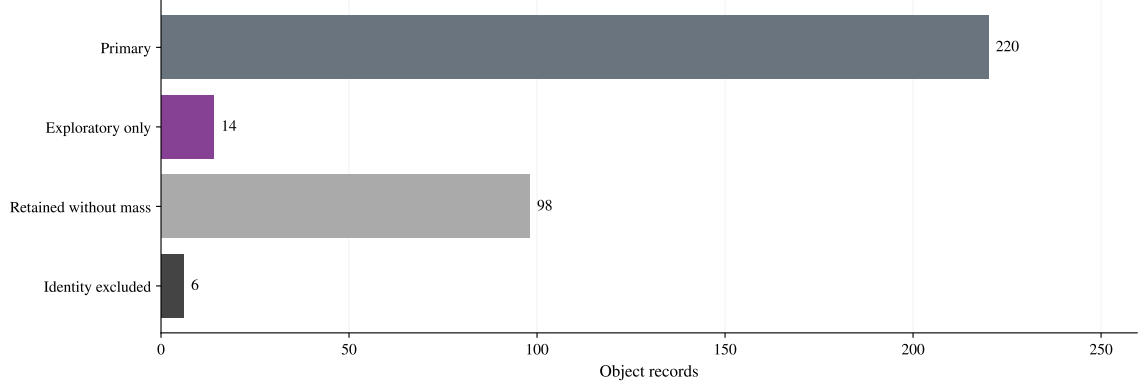


Figure 5: Sample accounting for 338 catalogue object records. Mutually exclusive categories comprise 220 primary objects, 14 additional exploratory objects, 98 retained without eligible masses, and 6 identity exclusions. The inclusive exploratory numerical sample contains 234 objects.

Table 8 gives every admitted source family. We count measurements and preferred objects separately. Each preferred object is counted once under its adopted source. Auxiliary coordinates are attributed in the registry to THRILS (T. A. Hutchison et al. 2025), UHZ1 spectroscopy (A. D. Goulding et al. 2023), and the versioned JADES DR3 GOODS-S prism catalogue. The machine-readable companion retains full selection-channel and caveat tags alongside source URLs. “Context” and “Limit” denote values retained outside numerical growth inference.

Table 8: Source-family inventory. Counts are admitted measurements / eligible mass measurements / preferred manuscript primary objects (M/G/P). The columns sum to 350/244/220; G includes alternatives, whereas P counts each selected object once. “Balmer” denotes single-epoch virial masses.

Source	M/G/P	Mass	Selection channel; principal caveat
C. J. Baccus & X. Xu (2026)	49/49/48	Balmer	NIRSpec broad lines; Frozen preprint masses; calibration
Ó. A. Chávez Ortiz et al. (2026)	1/0/0	Context	UV/optical diagnostics; Model-dependent mass
J. Chisholm et al. (2024)	1/0/0	None	High-ionization lines; No canonical mass
K. Davis et al. (2026) / THRILS	6/6/6	Balmer	EELG-selected spectroscopy; Virial calibration
Q. Fei et al. (2026) / GLIMPSE	10/10/8	Balmer	Lensed broad lines; Lensing corrections
J. E. Greene et al. (2024) / UNCOVER	9/9/9	Balmer	Lensed broad lines; Lens model; BLR interpretation
Y. Harikane et al. (2023)	10/10/5	Balmer	NIRSpec broad lines; Local virial calibration
I. Juodžbalis et al. (2026a) / JADES	23/23/17	Balmer	Broad Balmer lines; Extinction; calibration
M. Killi et al. (2024)	1/1/1	Balmer	Broad H α ; Linear mass-error conversion
D. D. Kocevski et al. (2025) / LRDs	6/6/6	Balmer	Broad Balmer lines; Dust correction; calibration

Table 8 continued

Source	M/G/P	Mass	Selection channel; principal caveat
R. L. Larson et al. (2023) / CEERS-1019	1/1/1	Balmer	Broad Hbeta; Broad-component significance
X. Lin et al. (2024) / ASPIRE	16/16/16	Balmer	Broad Halpha; Virial calibration
X. Lin et al. (2026) / COSMOS-3D	13/13/13	Balmer	NIRCam grism broad lines; Possible scattering bias
J. Lyu et al. (2024) / SMILES	19/0/0	None	MIRI SED selection; SED-dependent AGN identification
R. Maiolino et al. (2024) / GN-z11	1/1/0	UV	UV lines; UV estimator outside primary
S. Mascia et al. (2026)	8/0/0	None	Compact blue broad emitters; No canonical mass
J. Matthee et al. (2024) / EIGER-FRESCO	20/20/19	Balmer	Broad Halpha; Dust; virial calibration
G. Mazzolari et al. (2025)	25/0/0	None	Narrow-line diagnostics; Star-formation overlap
G. C. K. Leung et al. (2026) / MEOW	12/0/0	None	MIRI SED selection; SED-dependent AGN fraction
R. P. Naidu et al. (2026) / MoM-BH*-1	1/0/0	Context	Broad Hbeta / SED; Scattering; stellar alternative
L. Napolitano et al. (2025a) / GHZ9	1/0/0	Context	UV lines / X-ray evidence; Mass assumes Eddington rate
L. Napolitano et al. (2025b) / GHZ4,7	2/0/0	None	Narrow / high-ionization lines; Tentative diagnostics
W. Ren et al. (2025) / ALPINE-CRISTAL	7/7/1	Balmer	Host-selected broad Halpha; Conditional BLR interpretations
J. Scholtz et al. (2025) / JADES	21/0/0	None	Narrow / high-ionization lines; Tentative classifications
D. D. Kocevski et al. (2026) / Skyfire (CEERS)	22/22/22	Balmer	Broad Halpha; Virial calibration
M. Tang et al. (2025)	3/0/0	None	High-ionization lines; Ionizing-source ambiguity
A. J. Taylor et al. (2025) / CEERS-RUBIES	37/37/35	Balmer	Broad Halpha; Slit loss; dust; calibration
H. Treiber et al. (2025) / UNCOVER	2/0/0	None	UV-line diagnostics; Stellar alternatives
H. Übler et al. (2024) / ZS7	1/1/1	Balmer	Offset broad Hbeta; Error includes calibration scatter
Á. Bogdán et al. (2024); F. Zou et al. (2026) / UHZ1	2/0/0	Context	X-rays / SED; Later detection disputed
Z. Zhang et al. (2026) / LRDs	5/0/0	Limit	Narrow lines / SED; Mass upper limits
M.-Y. Zhuang et al. (2026) / NEXUS	15/12/12	Balmer	NIRCam WFSS broad lines; Twelve masses lack errors

A.1 Preferred measurements and identity exclusions

The measurement–object link table records the source, publication version, quoted uncertainty, and reason for each preferred measurement. The 3 unresolved identity groups are Baccus GDS_1210_9515 versus JADES GS-8083, GS-10013704 versus Scholtz 99671, and Scholtz 16745 versus 208643. All 6 records are excluded from manuscript inference. Three have eligible masses and otherwise pass the primary cuts; the other 3 lack eligible masses. Their spectra, images, and apertures require reconciliation before readmission. Detailed admission and revision records accompany the catalogue release.

Appendix B Growth-model derivations and implementation

B.1 Accretion and exponential growth

The growth constraints follow from the relation between accretion luminosity, retained mass, and available cosmic time. For electron-scattering opacity in ionized hydrogen, the Eddington luminosity and luminosity-based Eddington ratio are

$$L_{\text{Edd}} = \frac{4\pi G M_{\text{BH}} m_p c}{\sigma_T}, \quad f_{\text{Edd}} = \frac{L_{\text{bol}}}{L_{\text{Edd}}}, \quad t_{\text{Edd}} = \frac{M_{\text{BH}} c^2}{L_{\text{Edd}}} = \frac{c \sigma_T}{4\pi G m_p} \simeq 0.45 \text{ Gyr}. \quad (4)$$

Here G , c , m_p , and σ_T denote the gravitational constant, speed of light, proton mass, and Thomson cross section. With radiative efficiency $0 < \epsilon < 1$ and rest-mass inflow rate $\dot{M}_{\text{in}}$,

$$L_{\text{bol}} = \epsilon \dot{M}_{\text{in}} c^2, \quad \dot{M}_{\text{BH}} = (1 - \epsilon) \dot{M}_{\text{in}} = \frac{1 - \epsilon}{\epsilon} \frac{f_{\text{Edd}}}{t_{\text{Edd}}} M_{\text{BH}}. \quad (5)$$

The inflow rate is defined at the radiating flow; earlier mass loss is omitted. Dividing Eq. 5 by M_{BH} and integrating at fixed ϵ gives

$$\ln\left(\frac{M_{\text{BH}}}{B_{\text{merge}} M_{\text{seed}}}\right) = \frac{1 - \epsilon}{\epsilon} \frac{1}{t_{\text{Edd}}} \int_{t(z_{\text{seed}})}^{t(z_{\text{obs}})} f_{\text{Edd}}(t) dt = \frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \Delta t}{t_{\text{Edd}}}. \quad (6)$$

Exponentiating yields Eq. 2; solving for the average rate and applying the non-negative floor yields Eq. 3. For a constant positive f_{Edd} , the e-folding time is

$$t_{\text{efold}} = \frac{\epsilon}{1 - \epsilon} \frac{t_{\text{Edd}}}{f_{\text{Edd}}}. \quad (7)$$

It is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\text{Edd}} = 1$.

B.2 Merger multiplier and inverse seed mass

The implementation evaluates Eq. 2 in log mass:

$$\log_{10}\left(\frac{M_{\text{BH}}}{M_{\odot}}\right) = \log_{10}\left(\frac{M_{\text{seed}}}{M_{\odot}}\right) + \log_{10} B_{\text{merge}} + \frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \Delta t}{t_{\text{Edd}} \ln 10}. \quad (8)$$

We evaluate $B_{\text{merge}} = 1$ (no boost) and 2 (a factor-of-two mass boost). At fixed accretion history,

$$\frac{M_{\text{BH}}(B_{\text{merge}} = 2)}{M_{\text{BH}}(B_{\text{merge}} = 1)} = 2, \quad \Delta \log_{10} M_{\text{BH}} = \log_{10} 2 \simeq 0.3010. \quad (9)$$

Solving the same relation for seed mass gives

$$\log_{10}\left(\frac{M_{\text{seed,req}}}{M_{\odot}}\right) = \log_{10}\left(\frac{M_{\text{BH}}}{M_{\odot}}\right) - \log_{10} B_{\text{merge}} - \frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \Delta t}{t_{\text{Edd}} \ln 10}. \quad (10)$$

Here $M_{\text{seed,req}}$ is the seed mass required for a specified average rate, efficiency, growth interval, and merger multiplier.

B.3 Cosmic time

Cosmic ages use the flat matter-plus-Lambda relation

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \operatorname{asinh} \left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}} (1+z)^{-3/2} \right], \quad (11)$$

with $H_0 = 67.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$. The Hubble constant is converted to inverse time; ages and growth intervals are in Gyr. Radiation is neglected.

B.4 Two-state duty cycle

For a two-state history with the same efficiency in both states,

$$\overline{f_{\text{Edd}}} = D f_{\text{burst}} + (1 - D) f_{\text{quiet}}, \quad D_{\text{req}} = \frac{\overline{f_{\text{Eddreq}}} - f_{\text{quiet}}}{f_{\text{burst}} - f_{\text{quiet}}}. \quad (12)$$

Here D is the active-state time fraction, $f_{\text{burst}} > f_{\text{quiet}} \geq 0$, and the diagnostic requires $\overline{f_{\text{Eddreq}}} \geq f_{\text{quiet}}$. We retain values $D_{\text{req}} > 1$ to flag histories that would require more active time than is available. The inferred duty cycle depends on the assumed burst and quiet-state rates.

B.5 Sampling, mass conversions, and track selection

Draws use seed 20260808 and a stable identifier-based random stream. For 10,000 draws, the binomial Monte Carlo standard error of a probability is at most 0.005 (approximately 0.0022 at probability 0.95). This numerical error measures the precision from a finite number of draws. The 12 mass-bearing NEXUS objects have no published mass errors: repeated point values produce zero-width stored intervals, while probabilities and uncertainty ranks remain unavailable. Table 2 identifies separately reported calibration scales and scatter already included in the quoted errors.

The mass-scale tests add the same fixed offset to every original mass draw. The final published Killi mass $8_{-0.4}^{+0.5} \times 10^8 M_\odot$ is converted by transforming both linear bounds, giving $\log_{10}(M/M_\odot) = 8.90309_{-0.02228}^{+0.02633}$.

For Figure 1, candidate average rates span 0.1–2.0 in steps of 0.1. For each seed mass and fixed efficiency, we retain the 2 rates with the most primary objects within 0.5 dex of either the $B = 1$ or $B = 2$ prediction, requiring at least 5 such objects. Ties favour smaller median absolute residuals, then smaller rates. Most objects selected by this 0.5-dex plotting tolerance lie at $z \simeq 4$ –6. Baseline rankings, duty-cycle diagnostics, and seed-redshift–mass maps retain $\epsilon = 0.1$. Scientific conclusions use the required-rate ranking.

Appendix C Compatibility across seed masses and accretion rates

C.1 Efficiency prescription

For the illustrative spin-dependent maps, the ideal Kerr thin-disk efficiency is the ISCO binding efficiency (J. M. Bardeen et al. 1972):

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3r_{\text{ISCO}}}}, \quad (13)$$

$$r_{\text{ISCO}} = 3 + Z_2 - \operatorname{sgn}(a) \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \quad (14)$$

$$Z_1 = 1 + (1 - a^2)^{1/3} [(1 + a)^{1/3} + (1 - a)^{1/3}], \quad Z_2 = \sqrt{3a^2 + Z_1^2}. \quad (15)$$

The dimensionless spin is $a = cJ/(GM_{\text{BH}}^2)$, where J is the signed angular momentum relative to the disk, and r_{ISCO} is in units of GM_{BH}/c^2 . The ideal efficiencies at $a = -1, 0, +1$ are approximately 0.038, 0.057, and 0.423, respectively, neglecting photon capture and stress at the ISCO. Logarithmic luminosity growth at supercritical inflow is motivated by advection and wind models (J. Poutanen et al. 2007). We use a simplified logarithmic relation with coefficient 1 and omit radial mass loss. For Eddington ratios above unity, the maps adopt

$$f_{\text{Edd}} = 1 + \ln \dot{m}, \quad \dot{m} = \frac{\dot{M}_{\text{in}}}{L_{\text{Edd}}/(\epsilon_a c^2)}, \quad \epsilon_{\text{eff}} = \begin{cases} \epsilon_a, & f_{\text{Edd}} \leq 1, \\ \epsilon_a f_{\text{Edd}} \exp(1 - f_{\text{Edd}}), & f_{\text{Edd}} > 1. \end{cases} \quad (16)$$

The logarithmic luminosity relation applies only for super-Eddington accretion. Each compatibility cell holds both the rate and its effective efficiency constant over the interval. The grid tests constant accretion states with fixed spin and omits relativistic slim-disc dynamics. For time-varying efficiency, growth depends on $\int [(1 - \epsilon(t))/\epsilon(t)] f_{\text{Edd}}(t) dt$, which requires the joint time dependence of efficiency and accretion rate.

Figure 6 gives the full scenario grid. A primordial interpretation of the broad seed-mass interval requires additional assumptions about seed formation. The descriptive fractions use point masses and the inclusive interval criterion in Section C.2, without error integration. The Supplementary Material discusses this interval for pre-existing primordial black holes and illustrates an earlier start of accretion.

C.2 Seed-mass ranges and compatible fractions

For each assumed accretion rate, efficiency, starting redshift, and merger contribution, we solve Eq. 10 for the initial seed mass that would grow to the observed black-hole mass. We then check whether that seed falls in each of 4 mass ranges: $10\text{--}100 M_{\odot}$ (light), $10^3\text{--}10^4 M_{\odot}$ (intermediate), $10^4\text{--}10^6 M_{\odot}$ (heavy), and $10^2\text{--}10^6 M_{\odot}$ (broad). We count an object as compatible with a range when some seed within it can reproduce the observed mass under these assumptions. The grid uses $z_{\text{seed}} = 30$, 3 spin/efficiency cases, merger multipliers 1 and 2, and average rates 0.3, 1, and 2. These maps use the supercritical efficiency prescription in Section C.1.

The seed-mass ranges overlap and are tested separately, without probability weights. The broad range spans several possible seed-formation channels. If the seed mass needed to reproduce an object is smaller than the lower bound of a tested range, even the smallest seed in that range would grow larger than the observed black hole at the assumed accretion rate. That object is therefore excluded from the count for that range and set of growth assumptions. For each set of assumptions, we report the percentage of catalogue objects whose observed masses can be reproduced from a seed in the tested range. These percentages describe this literature sample. Estimating the corresponding percentages among all high-redshift black holes would require accounting for how each contributing study selected and detected its objects.

The primary and exploratory samples show similar compatibility patterns (Fig. 6). For heavy seeds at $a = +1$ and $f = 1$, changing B_{merge} from 1 to 2 raises compatibility from 58 to 71 per cent in the primary sample (59 to 71 per cent inclusive). At higher accretion rates, even the smallest seed in a tested mass range can grow larger than the observed black hole. Such objects reduce the percentage reproduced from seeds in that range. These fractions use central mass estimates and the efficiency prescription in Section C.1.

Compatibility across growth assumptions

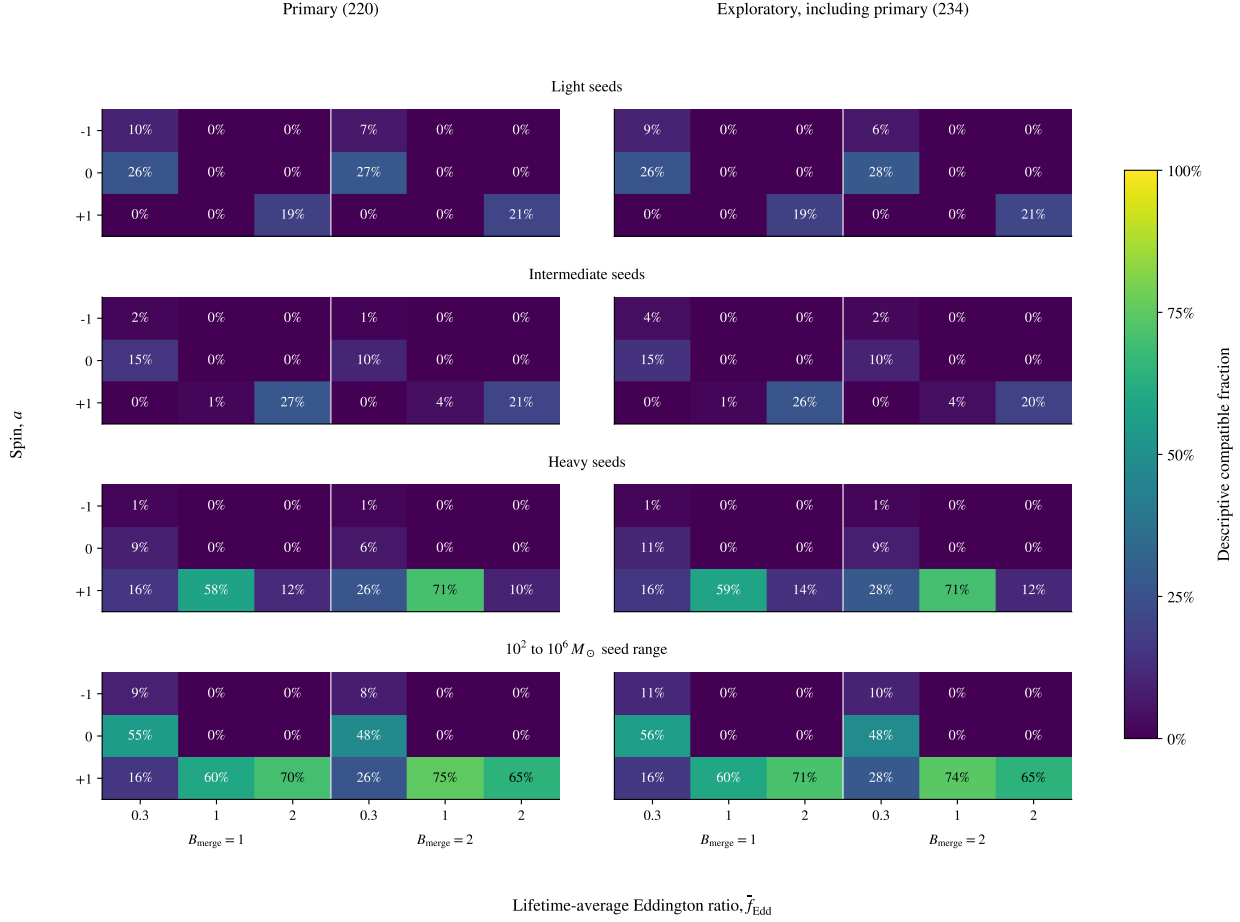


Figure 6: Constant-state compatibility fractions in the primary sample (left) and the inclusive exploratory sample (right). The exploratory sample includes all primary objects. Each column specifies merger multiplier B and fixed luminosity ratio f ; rows specify ideal spin-dependent efficiency, with the supercritical prescription in Section C.1. Seed intervals overlap and have no assigned probability weights.

Appendix D Seed mass and starting redshift

Figure 7 shows the required average Eddington ratio as a function of seed mass and starting redshift for UNCOVER-20466 and GN-z11. The common axes show how delaying growth increases the required rate and how a larger seed reduces it. The comparison extends the 2 starting redshifts used in Figure 3 over the interval $11 \leq z_{\text{seed}} \leq 30$. The full per-object parameter-map galleries are available online.

Appendix E Data and code availability

The catalogue, source registries, code, and manuscript analysis are available at <https://github.com/nniickels/highz-accretion-atlas>. The frozen v3 catalogue baseline is identified by Git commit `a40a0d2`; version manifests under `releases/` record the dataset artifact hashes. Comparison results and tables are in `paper/analysis/`. External inputs are in `paper/review_inputs.json`. An archival DOI is pending.

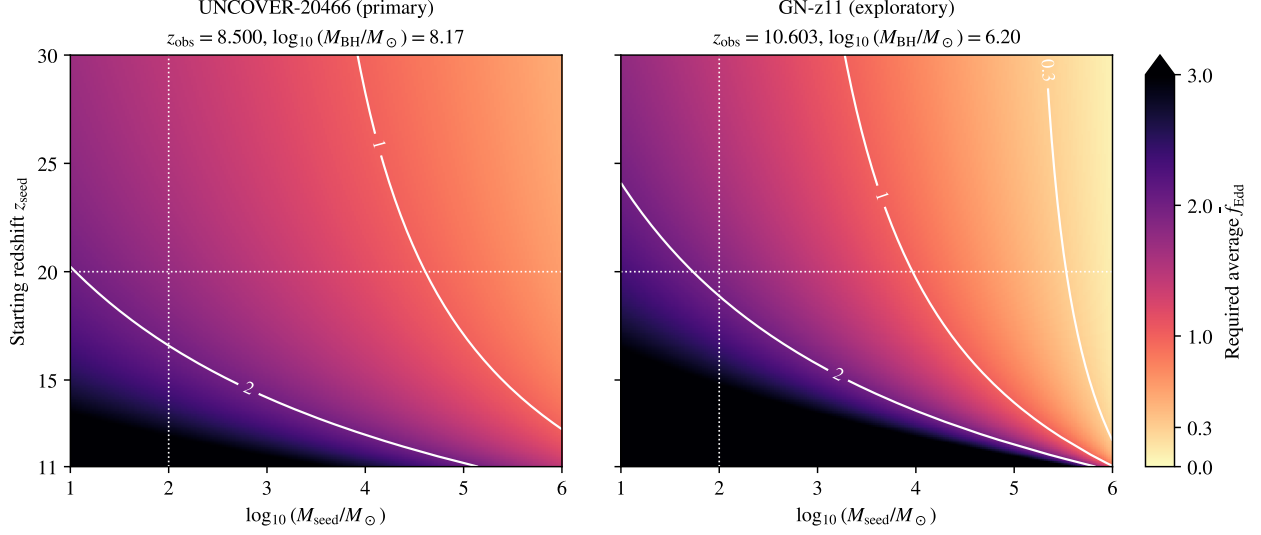


Figure 7: Required average Eddington ratio for UNCOVER-20466 (primary sample) and GN-z11 (exploratory sample), using their adopted central masses and redshifts shown above each panel. Both panels use $\epsilon = 0.1$, no merger boost, and the same axes and colour scale. White contours mark required ratios of 0.3, 1, and 2; dotted guides mark a $100 M_{\odot}$ seed and $z_{\text{seed}} = 20$. Colours saturate above a required ratio of 3. Mass uncertainties are not propagated in these maps.

The numbered notebooks reproduce the catalogue, analysis, figures, and parameter-map galleries using pinned dependencies. Instructions and verification procedures are in `docs/guides/reproducibility.md`. The source-family inventory in Appendix A links measurements, methods, selection channels, and major caveats. Full identifiers, source-native observables, alternative measurements, and per-object maps remain available.

These checks verify transcription and numerical reproduction. The validity of the mass estimators and unresolved object identities require observational assessment. All records in the 3 unresolved groups listed in Section 5.4 are excluded from manuscript inference.

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